R-INVARIANT THICKENING DECOMPOSITION OF U FOR A GROUPOID SCHEME WITH FINITE SOURCE AND TARGET, WITH FINITE LOCALLY FREE RESTRICTION ON THE CLOSED PART

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Allowed standard prerequisites (editorial).

- The finite stratification of U for a groupoid scheme with finite source and target: U = Z_0 superset Z_1 superset Z_2 superset ... R-invariant closed subschemes with empty intersection, whose strata differences carry restrictions with finite locally free source and target of rank r (more-groupoids chapter, Lemma lemma-finite-stratify, canonical tag 0AB9). cited at proof lines 2686-2687, 2735-2736 and 2739. Supplies the stratification that drives the construction, the rank r finite locally free property (5) on the new stratum, and the empty intersection that yields the covering property (2).

Source reference. Original lemma 0AB9: sources/stacks-chow/tags.txt, source lines 13012–13012. Cited at proof lines 2686-2687 for the stratification U = Z_0 superset Z_1 superset ... that drives the induction, at lines 2735-2736 for the rank r finite locally free property (5) on the new stratum, and at line 2739 for the empty intersection of the strata giving the covering (2).

Lemma 0.2. *Let (U, R, s, t, c) be a groupoid scheme over a scheme S. Assume s, t are finite. There exists an open subscheme $W \subset U$ and a closed subscheme $W' \subset W$ such that*

(1) *W and W' are R-invariant,*
(2) *$U = t(s^{-1}(\overline{W}))$ set theoretically,*
(3) *W is a thickening of W', and*
(4) *the maps s', t' of the restriction (W', R', s', t', c') are finite locally free.*

Proof. Consider the stratification $U = Z_0 \supset Z_1 \supset Z_2 \supset \ldots$ of Lemma 0AB9.

We will construct disjoint unions $W = \coprod_{r \geq 1} W_r$ and $W' = \coprod_{r \geq 1} W'_r$ with each $W'_r \to W_r$ a thickening of R-invariant subschemes of U such that the morphisms s'_r, t'_r of the restrictions $(W'_r, R'_r, s'_r, t'_r, c'_r)$ are finite locally free of rank r. To begin we set $W_1 = W'_1 = U \backslash Z_1$. This is an R-invariant open subscheme of U, it is true that

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W_0 is a thickening of W'_0, and the maps s'_1, t'_1 of the restriction $(W'_1, R'_1, s'_1, t'_1, c'_1)$ are isomorphisms, i.e., finite locally free of rank 1. Moreover, every point of $U \setminus Z_1$ is in $t(s^{-1}(\overline{W_1}))$.

Assume we have found subschemes $W'_r \subset W_r \subset U$ for $r \leq n$ such that

(1) $W_1, \ldots, W_n$ are disjoint,
(2) W_r and W'_r are R-invariant,
(3) $U \setminus Z_n \subset \bigcup_{r \leq n} t(s^{-1}(\overline{W_r}))$ set theoretically,
(4) W_r is a thickening of W'_r,
(5) the maps s'_r, t'_r of the restriction $(W'_r, R'_r, s'_r, t'_r, c'_r)$ are finite locally free of rank r.

Then we set

$$W_{n+1} = Z_n \setminus \Big(Z_{n+1} \cup \bigcup\nolimits_{r \leq n} t(s^{-1}(\overline{W_r}))\Big)$$

set theoretically and

$$W'_{n+1} = Z_n \setminus \Big(Z_{n+1} \cup \bigcup\nolimits_{r \leq n} t(s^{-1}(\overline{W_r}))\Big)$$

scheme theoretically. Then W_{n+1} is an R-invariant open subscheme of U because $Z_{n+1} \setminus \overline{U \setminus Z_{n+1}}$ is open in U and $\overline{U \setminus Z_{n+1}}$ is contained in the closed subset $\bigcup_{r \leq n} t(s^{-1}(\overline{W_r}))$ we are removing by property (3) and the fact that t is a closed morphism. It is clear that W'_{n+1} is a closed subscheme of W_{n+1} with the same underlying topological space. Finally, properties (1), (2) and (3) are clear and property (5) follows from Lemma 0AB9.

By Lemma 0AB9 we have $\bigcap Z_r = \emptyset$. Hence every point of U is contained in $U \setminus Z_n$ for some n. Thus we see that $U = \bigcup_{r \geq 1} t(s^{-1}(\overline{W_r}))$ set theoretically and we see that (2) holds. Thus $W' \subset W$ satisfy (1), (2), (3), and (4). $\square$